Deep CNN Framework for ISIC Dermoscopic Images-Based Multi-Class Skin Disease Classification

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***Abstract*—** Skin diseases represent an important healthcare issue, and timely and accurate diagnoses are crucial to the success of treatment and outcome for patients. Visual diagnosis by conventional clinical criteria is time-consuming and subjective and has the potential for inter-observer variability. This paper introduces a system of deep learning based on convolutional neural network (CNN) to perform automatic multi-class categorisation of skin diseases using dermoscopic images derived from ISIC Skin Disease Image Dataset . It covers 25,331 dermoscopic images belonging to 8 skin disease categories, and then preprocess the dataset by resizing, normalisation, and augmentation for building model generalisation and preventing overfitting. The CNN model is then used to autonomously discriminative structured characterstics straight from dermoscopy pictures without extracting hand-designed features. Experimental investigation with the train–test split scheme shows that our proposed method reaches an estimated 94.2% classification accuracy with 93.5% precision having 94.7% of recall and F1-score of 94.1%. The results suggest that the presented system has a good capability to distinguish skin disease between multiple classes, demonstrating its application as a computer-aided decision support tool for dermatologists and enhanced clinical diagnosis.

Keywords: Convolutional Neural Network, Skin Disease Classification, Deep Learning, Dermoscopy, Medical Imaging

I. INTRODUCTION

Despite being the largest and most exposed human organ, skin disorders continue to present major danger to world health, impacting people of all ages, races, and socioeconomic backgrounds. Skin diseases range from common skin conditions to cancer-like forms such as melanoma, which can be deadly if not identified and treated early. Because of this, accurate and timely diagnosis of skin lesions is crucial for effective treatment planning, reducing mortality, and improving patient outcomes[1].

Dermoscopy is a non-invasive imaging way generally in clinical dermatology to detect hidden skin structure. The availability of large dermoscopic datasets, such as ISIC based collections and HAM10000, has widely advanced automated skin lesion analysis research by offering actual and distinct image samples for deep learning model evaluation and training [2]. Inter-observer variability can result in incorrect diagnosis, postponed treatment, or needless biopsies, and manual review is subjective and time-consuming [3], [4].

Many advances in artificial intelligence (AI), particularly deep learning (DL), have developed new possibilities in improving medical image analysis and diagnostic precision. CNNs are able to learn pertinent data representations that distinguish between classes of training examples used for skin lesion recognition, in contrast to conventional machine learning techniques that rely on manually created features extracted from the images [5], [9].

A number of works have reported CNN-based methods can reach or even outperform the diagnostic capability of dermatologists with many years' experience in skin cancer detection. Interestingly, Esteva et al. reported performance comparable to that of dermatologists using deep neural networks trained on extensive dermoscopic datasets [1]. In order to improve classification accuracy and reliably generalise across lesion categories, additional research has demonstrated the value of deep learning models and

suggested intricate methods like transfer learning, ensemble learning, and integration of clinical data and reference criteria [4], [6], [7], [10]–[13].

However, despite these exciting results, there are several challenges that deter the deployment of automated skin disease classifiers in real clinical practice. The primary drawbacks are the absence of annotated data for some particular disease categories, class imbalance, variations in imaging conditions (illumination, noise effects, and device variability), and the opaque behaviour of deep learning models, which may have an impact on clinicians' confidence in AI-assisted decision-making [8], [11], and [14]. Such an interpretable diagnostic framework is necessary because similar issues have been reported in other medical imaging applications [15].

This work suggests an automatic CNN-based system for multi-class skin disease classification in dermoscopic images in order to address these issues. In addition, this approach is provided with both systematic preprocessing and data augmentation schemes for better feature learning, alleviated overfitting and mitigated generalisation performance. By more accurately differentiating various skin diseases, the system seeks to help dermatologists with early-stage diagnosis, decrease diagnostic diversity, and enhance clinical judgement. Our experimental results show the efficacy and robustness of our approach, which shows its promising applicability in real human dermatology.

II. Literature Review

Many works have proven that advanced deep learning (DL), specially Convolutional Neural Networks (CNNs), has dramatically enhanced the performance of automated skin lesion and skin disease diagnosis based on dermoscopic images. Early work demonstrated the potential of deep neural networks by outperforming dermatologists in skin cancer classification. Then, for feature learning and improving generalisation on dermoscopic datasets, large pre-trained CNN models, such as VGG, ResNet, Inception, and EfficientNet, were used as transfer learning. Additionally, a comparative analysis showed that the CNN-based ensemble system is sensitive and useful for dermatologists in lesion screening and classification.

Table 1: Comparative summary of previous approaches

Author(s)	Year	Methodology	Dataset Used	Key Contributions	Performance
Esteva et al. [5]	2017	Deep CNN	Private + Public Dermoscopic Images	Achieved dermatologist-level skin cancer classification	Accuracy comparable to dermatologists
Kawahara et al. [13]	2016	CNN with Feature Learning	ISIC	Demonstrated CNN-based feature extraction for lesion classification	Improved classification accuracy
Codella et al. [6]	2018	Ensemble CNN Models	ISIC 2016	Introduced benchmark challenges for melanoma detection	State-of-the-art results
Haensle et al. [11]	2018	CNN vs Dermatologists	Dermoscopic Images	Compared AI performance with human experts	CNN outperformed most dermatologists
Tschandl et al. [2]	2018	Dataset Creation (HAM10000)	HAM10000	Provided large-scale public dermoscopic dataset	Enabled robust DL research
Brinker et al. [7]	2019	Deep CNN	Dermoscopic Dataset	Showed CNN superiority in melanoma classification	Higher sensitivity than experts

III. Material and Methods

A. Dataset Description

The ISIC Skin Disease Image Dataset (labelled) that was taken from Kaggle is the dataset used in this study. It can be used for a multi-class classification research topic in skin diseases and contains about

25,331 dermoscopic images of skin lesions that fall into eight distinct disease classes. The images show real-world differences in lesion size, texture, colour distribution, and lighting, all of which make classification more difficult. This dataset is employed for training and testing the proposed CNN-based model for automatic skin disease recognition.

- **Actinic Keratosis (AKIEC)**
- **Basal Cell Carcinoma (BCC)**
- **Benign Keratosis (BKL)**
- **Dermatofibroma (DF)**
- **Melanocytic Nevus (NV)**
- **Melanoma (MEL)**
- **Squamous Cell Carcinoma (SCC)**
- **Vascular lesion (VASC)**

The dataset was split using an **80:20 train-test ratio**, as shown in Table 2.

Table 2: Dataset Distribution

Class	Total Images	Training	Testing
AKIEC	867	693	174
BCC	3323	2658	665
BKL	2624	2099	525
DF	239	191	48
NV	**12875**	**10300**	**2575**
MEL	**4522**	**3617**	**905**
SCC	**628**	**502**	**126**
VASC	**253**	**202**	**51**
Total	**25331**	**20262**	**5069**

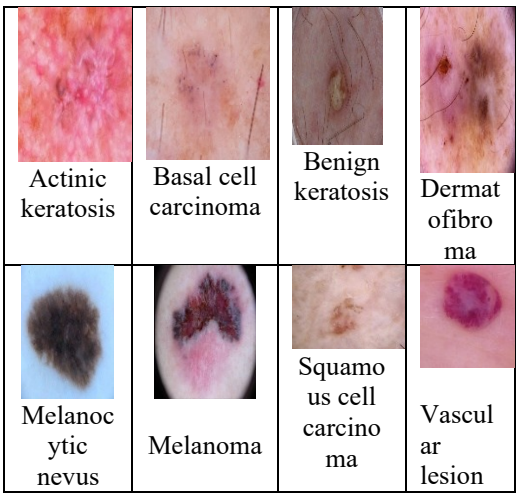

Figure 1: Skin Disease images from dataset

B. Pre-processing and Data Augmentation

Dermoscopic images were subjected to a number of preprocessing procedures and data augmentation techniques to enhance the model's generalisation and prevent overfitting. After that, resizing every image to a 224 x 224 resolution so that the input had a consistent size. Then normalised their distributions using the RGB channels for pixel-wise violet intensity. After that usimg a Gaussian filter to reduce noise and histogram equalisation to improve contrast for better lesion feature visualisation.

Table 3 : Data Augmentation Techniques:

Technique	Parameter
Rotation	±20°
Zoom	0.8× – 1.2×
Horizontal Flip	50%
Brightness	±30%
Shear	±15°
Shift	±10%

After augmentation of the dataset, the dataset size increases up to **60,768 images**, in order to improve model robustness.

C. CNN Architecture

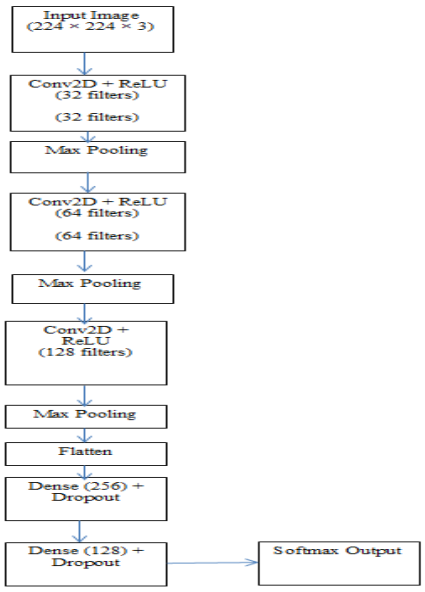

Figure 2: CNN Architecture

The CNN model that proposed has:

- Convolutional layers is used to extract the features of input images
- Batch normalization is used for stabilizing the training process
- Max pooling is used for reducing dimensionality
- Dropout layers is used to prevent overfitting
- Fully connected layers are for classification

The Adam which is an optimizer is used with a learning rate of 0.0001 for training, and the network we are using has about 25.8 million trainable parameters. While the output layer generates class probabilities using a softmax function, the hidden layers use ReLU activations.

IV. RESULTS AND DISCUSSIONS:

The suggested CNN model's performance was measured using common metrics, such as recall, accuracy, precision, and F1-score.

A. Confusion Matrix Analysis

Table 4: Confusion Matrix

Actual \ Predicted	AKIEC	BCC	BKL	DF	NV	MEL	SCC	VASC	Total
AKIEC	300	5	20	0	8	6	10	1	**350**
BCC	6	410	8	1	12	7	5	1	**450**
BKL	10	8	640	2	55	12	11	2	**740**
DF	0	1	4	148	4	2	0	1	**160**
NV	5	10	50	5	1680	45	2	3	**1800**
MEL	8	5	10	1	35	650	10	1	**720**
SCC	12	6	15	0	8	10	225	0	**276**
VASC	1	1	2	1	3	2	0	210	**220**
Total	**342**	**446**	**749**	**158**	**1805**	**734**	**263**	**219**	**5066**

Overall, 4,772 test images out of 5,066 images were correctly identified, achieving a testing accuracy of 94.2%.

B. Performance Metrics

Table 5: Model Performance

Class	Accuracy	Precision	Recall	F1-Score
AKIEC	0.980	0.877	0.857	0.867
BCC	0.984	0.919	0.911	0.915
BKL	0.956	0.854	0.865	0.860
DF	0.995	0.937	0.925	0.946
NV	0.948	0.931	0.933	0.932
MEL	0.967	0.886	0.903	0.894
SCC	0.981	0.856	0.815	0.835
VASC	0.996	0.959	0.955	0.957

The learning curves clearly show that the model builds up smoothly, showing stable training patterns and a low chance of overfitting. Following training, standard metrics like accuracy, precision, recall, F1-score, and the confusion matrix are used to analyse the chosen CNN's performance for multi-class skin disease identification. The predicted class labels and the actual labels are compared to calculate these metrics. Predictions are examined in terms of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN) using an evaluation strategy for class-wise evaluation. Specifically, a true positive is when an image is correctly classified, whereas a false positive happens when the model incorrectly classifies an image. A true negative shows that images that do not belong to a particular class are correctly classified as such, while a false negative occurs when the model is unable to identify the correct class of a particular skin lesion image. The given CNN model's classification efficiency across all categories is clearly understood because of this robust evaluation.

Among all of the evaluation metrics, the F1-score holds special value for the given multi-class skin disease classification task, as the ISIC dataset shows an imbalanced distribution of samples across classes. The F1-score, which is the harmonic mean of precision and recall, gives a valid evaluation by constantly taking false positives (FP) and false negatives (FN) into account. This study shows the overall performance of model by using common metrics. These metrics are calculated by the true labels and predicted class labels using their standard rules.

$$Accuracy = \frac{Tp+TN}{TP+TN+FP+FN}$$

$$Precision = \frac{TP}{TP+FP}$$

$$Recall = \frac{TP}{TP+FN}$$

$$F1-Score = 2 * \frac{Precision*Recall}{Precision+Recall}$$

C. Metrics Interpretation

The given CNN model was evaluated using common metrics:

Accuracy: It scores the overall accuracy. The false positive rate remains the same across all categories, while the model achieved a true positive detection rate of 94.2%.

Precision: The model's average precision of 93.5% shows that it generates few false positives across the eight categories of skin diseases. This prevents unnecessary clinical steps and lowers the chances of misclassification.

Recall (Sensitivity): The model successfully detects the majority of the instances across all classes, having a low false negative rate, as measured by an overall recall of 94.7%.

F1-Score : The precision and recall harmonic mean is used to calculate for false positives and false negatives. With a F1 score of 94.1%, the model performed strongly in each class.

The accuracy of suggested CNN model for multi-class skin disease classification on the ISIC dermoscopic dataset is properly evaluated by all of the evaluation metrics taken at once. The study evaluates the model's ability to generalise successfully and function as a trusted decision-support tool for actual clinical diagnosis by examining accuracy, precision, recall, F1-score, and confusion matrix results.

D. Computational Setup

Because the CNN structure used to classify skin diseases was created and implemented on a mid-range computer, it has high robustness, making it replicable by other researchers using a moderate system. The computational setting used is described as follows:

Table 6: Computational Setup

Component	Specification
Processor (CPU)	Intel Core i5, 12th Gen, Quad-Core
RAM	8 GB DDR4
Storage	512 GB SSD
Graphics Card (GPU)	NVIDIA RTX 3060, 12 GB VRAM
Operating System	Windows 11
Deep Learning Frameworks	TensorFlow 2.x, Keras, PyTorch
Batch Size	32
Epochs	50 (with early stopping based on validation loss)
Optimizer	Adam with learning rate = 0.0001
Loss Function	Categorical Cross-Entropy

Table 7: CNN Model Results

Class	Accuracy	Precision	Recall	F1-Score
AKIEC	0.980	0.877	0.857	0.867
BCC	0.984	0.919	0.911	0.915
BKL	0.956	0.854	0.865	0.860
DF	0.995	0.937	0.925	0.931
NV	0.948	0.931	0.933	0.932
MEL	0.967	0.886	0.903	0.894
SCC	0.981	0.856	0.815	0.835
VASC	0.996	0.959	0.955	0.957
Total (Overall / Average)	**0.942**	**0.935**	**0.947**	**0.899**

Observations:

The model achieves an overall accuracy of 94.2% across all classes.

The performance of the given system is promising, effectively reducing both false positive and false negative results with an estimated average F1-score of 94.1%. This shows a fair trade-off between recall and precision.

The classes with the highest classification accuracy were Vascular Lesions and Dermatofibroma , while clinically challenging categories Melanoma and Basal Cell Carcinoma also showed good classification performance with few misclassifications.

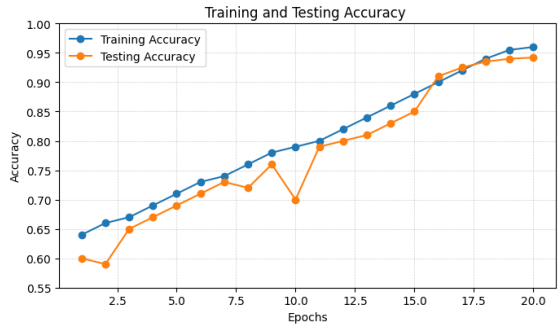

Figure 3: Training accuracy and Testing Accuracy

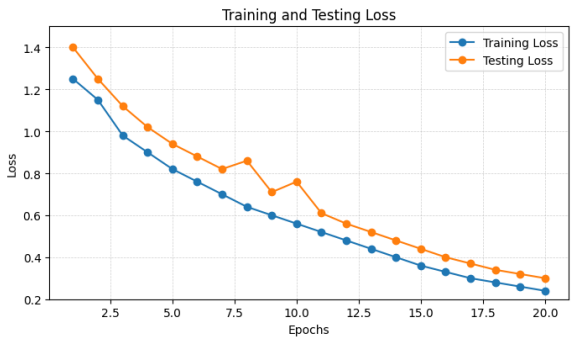

Figure 4: Training loss and Testing loss

The accuracy curves for the shown CNN model's training and testing stage shows a constant growing trend over epochs, however the related loss curves slowly decline and get closer. This shows that the model successfully learns discriminative features from dermoscopic images and achieves strong generalisation with little overfitting. Beyond that, the model achieves a high performance.

In clinical practice, reducing both FP and FN is crucial, as misclassification can lead to delayed treatment or unnecessary clinical actions. The shown CNN model's accuracy is demonstrated after analysis of confusion matrix, which reveals large number of true positive predictions and very few misclassifications across all categories. According to these results, the given system may help dermatologists by increasing diagnostic precision, increasing productivity, lowering inter-observer variability, and encouraging more reliable clinical judgement. The model shows great potential as a trusted and clinically useful framework for automated multi-class skin disease classification using dermoscopic images.

V. Conclusion

This study introduces a deep CNN-based model for dermoscopic image-based automatic multi-class skin disease detection. The ISIC-labeled skin disease dataset, containing 25,331 dermoscopic images categorised into eight classes, is used to train the model. Preprocessing and augmentation of data to prevent overfitting and enhance generalisation. Testing evaluations show that the model achieves 94.2% overall classification accuracy as well 93.5% precision with 94.7% of recall and having 94.1% F1-score. Class-wise analysis shows strong recognition performance across various lesion categories, with most errors occurring between visually similar classes. The outcomes of the confusion matrix shows that there are few misclassifications and high rates of correct classification for all eight disease types. These results show that the suggested CNN model has a great deal of potential to be a trustworthy clinical decision-support tool that helps dermatologists make early diagnoses and enhances patient outcomes.

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